

Reconstructing the SLRTP2025 Back-Translation Evaluator from Public Artifacts: What Reproduces, and What Does Not

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Abstract

We audit whether the released SLRTP2025 back-translation (BT) evaluator for sign language production can be reconstructed from its public artifacts, an evaluation-only repository without training scripts.

A line-level audit of our own reconstruction code identified seven defects relative to the released configuration’s semantics, including a validation-mask inversion, a loss-normalization mismatch (token-mean versus the released batch semantics, pinned arithmetically against the released training log), and a stopping-rule mismatch (the released log provably did not use patience-based early stopping). A faithful re-implementation that corrects all seven defects, with the released seed 42 as the primary run, recovers most of the evaluator’s in-domain competence: across eight seeds, dev BLEU-4 11.59–12.62 versus the released 13.38, with test-split decoding up to 13.11 versus the released 12.78. Two properties do not reproduce: the training-pool readout signature (released 78.8 BLEU / 70.7% exact match; faithful family 11.45–12.08 / 2.3–2.5%), and the response to a constructed adversarial replay probe, a post-selected estimate of **+10.24** sacreBLEU on the released evaluator. All 69 non-degenerate constructible evaluators, including the eight faithful runs at near-matched test competence, give negative probe gaps (range [−**2.80**, −**0.21**]).

The probe response is confirmed on a never-explored dev split (**+8.10**) and is entirely dependent on selecting donors with the scored reference text: pose-based reference-free retrieval collapses to **−12.33**, indistinguishable from random donors. Reference replacement with human back-translations is associated with a gap reduction to **+0.95** from **+10.03**—a single-annotator reference-frame intervention contrast—and pool-size-controlled donor-pool resampling leaves a **+8.38** residual origin contrast (95% conditional simulation interval [**+7.44**, **+9.34**]). These findings delimit what the public artifacts reproduce: surface competence, yes; the released evaluator’s training-pool memorization signature and its probe

response, no. We frame the probe throughout as an adversarial lexical-coupling stress test, not a general evaluator finding.

Keywords: sign language production, back-translation evaluation, reproducibility audit, reference-frame sensitivity, learned evaluation metrics, PHOENIX-2014T

1 Introduction

Sign language production (SLP) systems are commonly evaluated by back-translation (BT): a trained sign recognizer decodes each output pose sequence, and a text metric compares the decoding with the reference. This measures recognizer-readable content through a learned measurement model, so system rankings may reflect both the evaluated outputs and the inductive biases of a particular evaluator checkpoint. The present audit is framed as a *public-artifact reconstruction attempt*: whether our implementation, using the released SLRTP2025 public artifacts, can reconstruct the released BT evaluator. We address two research questions:

- RQ1. Can a config-exact, defect-free implementation of the released training configuration reconstruct the released evaluator’s competence and training-pool readout?
- RQ2. How sensitive is the constructed probe’s response to the reference text and to donor origin?

Beyond the two research questions, we report a *descriptive observation* (not a research question): the probe response does not reproduce under any constructible evaluator. This includes eight faithful runs whose test-split decoding competence matches or exceeds the released evaluator’s (up to 13.11 vs. 12.78), so the non-reproduction is no longer attributable to uniformly lower surface competence; what the faithful runs still lack is the released evaluator’s training-pool memorization signature (78.8 BLEU / 70.7% exact match vs. at most 12.08 / 2.5%). Across all constructible evaluators the probe response instead tracks that signature, which no public artifact documents how to produce. The replay probe itself is an *adversarial lexical-coupling stress test*: the source-conditioned retriever selects donors by surface vocabulary overlap with the German source, which is identical to the scoring reference in PHX-public, so the probe deliberately targets the strongest form of the shortcut rather than a deployable retrieval pipeline.

Prior work shows BT scores can be modest for recorded signing and can improve without faithful target conditioning Walsh et al. (2024); Hong and Košecká, Jana (2026) (Hong and Košecká, arXiv preprint, preliminary); sign-native or pose-aware metrics provide complementary evidence Kim et al. (2024); Imai et al. (2025); Jiang et al. (2025), yet BT remains central to SLRTP2025 Walsh et al. (2025). Chen et al. Chen et al. (2022) showed that learned BERT-based evaluation metrics can fail to reproduce due to undocumented preprocessing, missing code, and baseline implementation differences—a methodology directly relevant to this audit, in which seven defects in our own reconstruction implementation were identified and then corrected in a faithful re-implementation. An unresolved question is whether a retrieval-reference

stress-test ordering persists across independently trained instances of the same BT architecture.

This question is exposed by an adversarial retrieval–reference probe on PHX-public: under the released evaluator, source-conditioned whole-donor retrieval scores 23.02 sacreBLEU (0–100 scale) versus 12.78 for the recorded test poses. Because the retriever selects donors by lexical overlap with the source sentence—which is identical to the scoring reference in PHX-public—this probe is a deliberately constructed adversarial stress test, not evidence of general evaluator preference or better signing.

We address the two research questions with the following evidence (full results in §4.2–§4.3). For RQ1, the faithful re-implementation recovers most of the released evaluator’s surface competence but not its training-pool readout. For RQ2, reference replacement with human back-translations is associated with a large gap reduction (a reference-frame intervention contrast), and donor-pool substitution reverses the sign. The source-conditioned retriever shares surface vocabulary with the original reference (§3.2), so the reference-frame result conflates reference-text properties, lexical-coupling disruption, paraphrase shift, and BLEU floor effects.

Why a constructed probe matters.

Retrieval-augmented SLP entries draw donors from the evaluator’s training pool, so the adversarial probe isolates a risk that production systems face in diluted form. The audit contributes a transferable template (provenance table, checkpoint registry with SHA-256 deduplication, evaluator-replication controls, teacher-forced-vs-free-decode diagnostics) for public-artifact reproducibility testing. The specific probe response is a PHX-public released-evaluator case study and does not support general claims about BT evaluation or retrieval-based SLP.

2 Related Work

Note on preprints. Where we cite 2025–2026 arXiv preprints Hong and Košecká, Jana (2026); Cory et al. (2026); Alkain et al. (2026); Czehmann et al. (2026); Artiaga et al. (2025); Desai et al. (2024), we treat their findings as preliminary reports rather than peer-reviewed results; conclusions in this audit do not hinge on the correctness of any single preprint. All arXiv citations were accessed on 2026-08-10.

Learned evaluation in sign language production.

Neural SLP often evaluates with a frozen sign recognizer Camgoz et al. (2018); Saunders et al. (2020); Saunders et al. (2021) proposed BT as an SLP evaluation protocol, arguing it shifts absolute scores but not rankings between similar configurations—a condition unmet in our family. SignBLEU Kim et al. (2024), SiLVERScore Imai et al. (2025), and newer human-correlated metrics Jiang et al. (2025); Cory et al. (2026) offer complementary views, but BT BLEU remains the *standardized infrastructure* on which the SLRTP2025 ranking depends Walsh et al. (2025); auditing its reproducibility is infrastructure maintenance, not a metric-superiority claim.

Retrieval and shortcut risks.

Retrieval-based SLP ranges from phonetic composition to retrieval-enhanced generation Walsh et al. (2024,?); Zuo et al. (2025); Tasyurek et al. (2025), and SLRTP2025’s leading entry is retrieval-based, making donor copying pertinent Walsh et al. (2025). Memorization results in retrieval-augmented language modeling show strong task scores can reflect reproduction of training examples Carlini et al. (2021); Kandpal et al. (2022); Khandelwal et al. (2020); a shortcut under one learned evaluator does not show the ordering persists under another checkpoint.

Robustness of learned metrics and corpus audits.

Hong and Košecká Hong and Košecká, Jana (2026) *report* (preprint) that BT-BLEU can diverge from target faithfulness; Walsh et al. Walsh et al. (2024) document a low recorded-pose BT baseline; Jiang et al. Jiang et al. (2025) recommend BT likelihood over decoded BLEU; and model-based generation metrics are further criticized by Yazdani et al. (2026); Cory et al. (2026); He et al. (2024). Two 2026 PHOENIX-2014T audits bear directly on our reversal: Czehmann et al. Czehmann et al. (2026) released human back-translations and documented reference information loss, Alkain et al. Alkain et al. (2026) showed the highest train–test text overlap among tested SLT corpora, and Artiaga et al. Artiaga et al. (2025) showed signer-dependent evaluation overestimates SLT capability. None runs multi-checkpoint evaluator experiments or donor-pool substitution controls.

Reproducibility of learned evaluation metrics.

Chen et al. Chen et al. (2022) showed that reproducing BERT-based evaluation metrics can fail due to undocumented preprocessing, missing code, and baseline implementation differences, even when model weights are released—a finding methodologically close to our public-recipe non-reproduction result. The difference is that our audit targets a full BT evaluator pipeline (architecture, training recipe, and checkpoint) rather than a metric alone, and the non-reproduction is demonstrated across 69 unique non-degenerate decoded runs (all 69 with strictly negative gaps; five degenerate excluded), corresponding to 68 distinct SHA-256 weight binaries after one validation-freq-misread/long-schedule seed-202 collision is deduplicated.

PHOENIX-2014T as a language resource: known limitations.

PHOENIX-2014T Camgoz et al. (2018) (8,257 weather-forecast interpretations by 9 signers) is the most widely used DGS parallel corpus, but four properties constrain our audit’s generalizability: (i) template density inflates train–test n-gram overlap Alkain et al. (2026) (32 items whose reference exactly matches a training text account for a disproportionate share of the gap); (ii) reference texts are speech transcriptions, not DGS translations Czehmann et al. (2026); (iii) the 178-keypoint pose materialization is a lossy signing representation: the 21 landmarks per hand and the 128-point face mesh encode joint centres and coarse facial geometry—including some finger articulation, palm-plane orientation, and mouth shape—but not appearance texture, hand–body contact, occlusion-robust fine depth, or reliable eye-gaze direction (our MSKA control uses 133 keypoints; SI Sup. L); (iv) signer-dependent evaluation can overestimate

capability Artiaga et al. (2025). The lexical coupling between source and reference amplifies these effects in our adversarial probe. The transferable audit components (provenance table, checkpoint registry, leakage controls) apply to any learned evaluator paired with a public training corpus.

3 Methods

3.1 Evaluation design

Let E_c denote a BT evaluator checkpoint, x_n the source sentence (German) for item n , and $o_{i,n}$ the pose sequence for output condition i and item n ; E_c produces $h_{c,i,n} = E_c(o_{i,n})$ and $B_{c,i} = M(\{(h_{c,i,n}, y_n)\}_{n=1}^N)$, where y_n is the reference text and M is the corpus-level sacreBLEU functional unless stated otherwise. We call a result a *descriptive non-reproduction* when a probe response observed under the released evaluator does not appear across independently constructed checkpoints under a fixed output set; the term is descriptive and does not imply a causal checkpoint-identity effect. The fixed primary set comprises recorded test poses (REC), random donor copy, source-conditioned whole-donor retrieval, a Progressive Transformer generator, and retrieval-generator compositions; oracle-gloss variants are diagnostic controls.

Operational definition of reconstruction competence.

We define an *author-operational* competence gate—not a public-sufficiency judgment—as dev BLEU-4 ≥ 12.38 and dev WER ≤ 86.37 (one-sided tolerance of ± 1 BLEU / ± 3 WER around the released evaluator’s 13.38 / 83.37). This gate is a sensitivity summary, not a community standard: the gate conclusion is insensitive across a $[0.5, 2.0]_{\text{BLEU}} \times [1.5, 4.5]_{\text{WER}}$ grid (§4.3), but other tolerance choices would give other pass/fail counts. The primary evidence is the *continuous* metric comparison itself (SI Sup. R: faithful dev BLEU-4 11.59–12.62 and train readout 11.45–12.08 vs. released 13.38 and 78.8); the gate merely summarises it for table presentation. Under the faithful family one of eight seeds passes the gate; the residual gap concentrates in the training-pool readout signature (§4.2).

The main evaluation uses PHX-public, a German weather-domain text-to-pose benchmark whose released SLRTP2025 pose records contain 7,060/515/641 sequences; one official test ID is absent from the released materialization, so every result is conditioned on the complete 641-record release (SI Sup. A). The three-way provenance table (official PHOENIX-2014T split, released pose materialization, and the released evaluator’s training manifest, confirmed via `config.yaml` and the shipped `.pt` tensors) is in SI Sup. P. Supplementary analyses use **PHX-holdout-MSKA** (133-joint, 1,538 windows) and CSL-Daily (Mode B, 1,176 sequences; SI Sup. L); the protocol summary is in SI Table S38.

All pose sequences are evaluated at 12.5 fps, matching the SLRTP2025 evaluator training distribution. The data flow is: official PHOENIX-2014T video at 25 fps $\rightarrow$ SLRTP2025 skeleton extraction at 25 fps $\rightarrow$ our `::2` subsampling to 12.5 fps for evaluator input; the official evaluation command uses `--fps 25` (the input file’s native rate), but the evaluator was trained on 12.5 fps-subsampled poses, so our subsampling matches the training distribution rather than the command-line flag. A three-path

regression confirms no double subsampling (details in SI Sup. F). PHX-public uses $\alpha=0.88$ (selected on dev); the supplementary splits use $\alpha=0.5$.

3.2 Systems and retrieval controls

For each target, source-conditioned whole-donor retrieval scans the complete training pool after Unicode NFKC normalization, whitespace normalization and lowercasing, selecting the sequence with maximum source-text token-set Jaccard similarity. Oracle-gloss retrieval instead uses target gloss and is treated as an input-leakage diagnostic; random retrieval selects a donor uniformly. Donor exclusion removes exact normalized text or gloss duplicates or donors above threshold τ ; gloss-based exclusion is therefore an offline diagnostic, not a deployable inference procedure.

Single canonical materialization. Every number in this paper is generated from one canonical donor registry built by the deterministic §3.2 algorithm (exact-normalized-text exclusion applied), whose SHA-256 is `9170a53026ab3263b451a3632a9e02-318745acabf12f0b679921477afbafa301`. Reproduction is supported at four layers (§ Sup. Q, Table S36): (i) **L1—recompute selected reported statistics**: a single command (`make core-audit`) regenerates the registry and verifies headline numbers from committed per-item decodes; it does *not* retrain models or decode from checkpoints, and does not verify every number in the paper; (ii) **L2—analysis re-execution**: each analysis script re-executes from the committed per-checkpoint data and hypotheses; (iii) **L3—model re-evaluation**: re-decoding fixed outputs from checkpoint weights requires the training bundle; (iv) **L4—from-scratch retraining**: full retraining requires the restricted PHOENIX-2014T licence, the exact environment, and approximately 200 GPU-hours. The `make core-audit` command executes L1 only and is not an end-to-end reproduction. A claim-to-file manifest mapping each headline number to its source file and verification command is in `claim_manifest.json` (artifact root). An earlier pre-exclusion materialization (PURE 23.79, gap +11.01) survives only as a sensitivity analysis in SI Sup. N.

Lexical coupling of the source-conditioned probe (adversarial design).

In PHX-public, the source sentence x_n (German weather report) is identical to the reference text y_n^{orig} at the sequence level (the reference is a speech transcription of the same German sentence). The TN-PURE-v1 retriever selects training-pool donors by token-set Jaccard similarity between x_n and training texts, so it preferentially retrieves donors whose accompanying training texts share surface-level vocabulary with both the source *and* the reference. Scoring a donor copy against this lexically coupled reference under a learned evaluator therefore confounds donor similarity, evaluator readout, and reference overlap. We explicitly frame this probe as an *adversarial stress test* targeting the strongest form of the lexical-coupling shortcut—not as evidence of general evaluator preference, deployable retrieval quality, or better signing. This framing has two consequences: (i) the +10.24 gap is the expected output of a deliberately adversarial design, not a discovery about the evaluator’s ranking behavior; (ii) the human-reference attenuation analysis (§4.3) reflects disruption of the lexical coupling as well as reference-text information loss, paraphrase shift, and BLEU floor effects—not a clean separation of any single mechanism.

Retrieval composition.

TN-PURE-v1 uses whole-donor copy with $[:,2]$ subsampling to 12.5 fps, preserving the donor’s original duration. TN-PTCOMP-v1 (a secondary control) composes a PT scaffold with donor hands and face via α -blending of upper-body joints ($\alpha=0.88$; full joint-set definitions in the artifact). The matched seen–unseen factorial retains 605/641 targets matched on duration and word count.

Donor-origin contrast retrieval systems.

To test whether the probe response co-occurs with donor origin, we construct frozen probe systems: **UNSEEN-PURE-v1** retrieves from the 640 other test poses; **CROSSFIT-PURE-A/B** split queries by hash into two folds; **SEEN-PURE-MATCHED-v1** selects a train-pool donor within ± 0.1 Jaccard of each UNSEEN donor (622/641 admit one). SEEN-PURE-v1 re-uses the canonical TN-PURE-v1 registry; these are constructed probes, not deployable generators (construction rules in the artifact).

3.3 Evaluator checkpoint family

The main inferential checkpoint family comprises the original frozen SLRTP2025 evaluator and 36 *recipe-conditioned* runs trained from scratch with the documented architecture on a common frozen partition: the eight *faithful* runs (§3.4; seeds 42–49), fourteen *partially-corrected* runs (4 validation-freq-misread + 10 step-corrected), and fourteen *legacy* runs (6 prospectively specified seeds 101–606 + 8 extension seeds 707–1405). The original bundle’s `validations.txt` logs 202 validation points over 2,828 optimizer steps (≈ 102 epochs), with decoded-BLEU selection reaching best dev BLEU-4 13.38 at step 1,820; the trajectory is analyzed in §4.3, and intermediate checkpoints are not published. All recipe-conditioned runs share the documented architecture (3L/256/512ff, 8 heads), tokenizer, manifests (7,060/515), $[:,2]$ subsampling, Adam (1e-3, wd .001), batch size 256, and plateau $\times 0.8$ LR schedule (full provenance in SI). In total 76 from-scratch training runs across 12 families (registry in SI Sup. D); 74 were beam-3 decoded on PHX-public under the canonical §3.2 donor registry, corresponding to 73 unique weight binaries — one validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact (same wandb run, identical per-step losses and best-checkpoint SHA-256). Of the 74 decoded training runs, five are degenerate (three $\alpha=1.0$ distillation students with empty hypotheses, two smallest ladder fractions with BLEU ≈ 0) and are excluded, so the non-replication result rests on 69 non-degenerate decoded training runs, all 69 with strictly negative gaps. The auto-generated accounting table is in SI Sup. P.

What the 76 runs are—and are not.

The 76 runs are *not* 76 same-recipe replication attempts. The SI family-provenance table (SI Sup. P) maps each family to the public source it draws on, the known deviation from the original training, the number of runs, the checkpoint-selection rule, and the resulting dev BLEU-4 range. The eight faithful runs implement the released config exactly (§3.4); the ten step-corrected and four validation-freq-misread runs are

partially corrected; the remaining 54 runs are diagnostic, sensitivity, or competence-extrapolation analyses. The public-recipe non-reproduction conclusion rests primarily on the faithful family, with the partially-corrected and legacy families as corroborating evidence.

Training-pool readout (descriptive observation).

We define *training-pool readout* as the in-sample autoregressive free-decode BLEU and exact-match (EM) rate on the full 7,060-item training set under beam-3 decoding—not teacher-forced accuracy. The released evaluator decodes its full 7,060-item training pool at free-decode BLEU 78.8 with 70.7% exact-match, while retaining dev/test performance of 13.38/12.78. The constructible family’s readout is far lower (faithful: 11.45–12.08 BLEU, 2.3–2.5% EM; reconstructions: 7.2–9.3 BLEU, 1.5–2.3% EM; distillation: 6.6–8.9 BLEU, 1.3–1.9% EM); no constructible checkpoint approaches both simultaneously. The training-pool free-decode BLEU ratio (released: $78.8 / 13.38 \approx 5.9$ vs. constructible family: 0.93–1.55) quantifies how much the model reads out above its in-domain performance. We report these as continuous descriptive observations: the released evaluator was observed first, the family range afterwards, and any threshold is post-hoc.

3.4 Reconstruction sanity checks and training diagnostics

Three checks on the recipe-conditioned families rule out basic plumbing failures (data loading, masking, decoding alignment, vocabulary mismatch, pose-subsample direction): a small-sample overfit test (50/100 sequences reach free-decode BLEU 100.00), a unified-inference regression (all checkpoints load through one pipeline with byte-identical vocabularies; the released checkpoint reproduces canonical REC 12.78 exactly), and the teacher-forced/free-decode split (SI Sup. R; the released evaluator’s 96.5% teacher-forced train token accuracy and 78.8 free-decode train BLEU are both far above the defective families’ 55–60% and 7.2–9.3). These checks do *not* rule out subtler training-objective, normalisation, or scheduler defects—indeed seven such defects were later found (§3.4)—so earlier rounds framed RQ1 strictly as “our disclosed reconstruction did not attain the released competence”, not “the public materials are insufficient”; that scope question is revisited in §5. Training logs confirm normal optimisation for the defective families (median train-NLL 5.83→2.09 over epochs 1–25; effective batch 256; a 25→12.5 fps subsampling regression test passes). Full diagnostics: `results/robustness_diagnostics.json`, `results/fps_regression_test.json` (SI Sup. R).

Seven implementation defects, and the faithful re-implementation.

A line-level audit of our own training scripts identified *seven* defects relative to the released `config.yaml`’s semantics. The first four were known from earlier rounds; the last three are newly identified here:

1. *CTC loss omitted* (v1) despite `recognition_loss_weight=1.0`; the ten step-corrected runs additionally trained with the recognition weight set to zero.

2. *Validation frequency misread* (v1): `validation_freq=14` is 14 optimiser *steps* (JoeyNMT semantics; the released log validates at steps 14, 28, ...), not 14 epochs.
3. *Selection branch never saved* (v1): the `--selection_bleu` branch had improved hard-coded `False`.
4. *Non-autoregressive selection decode* (v1.5/v2): selection BLEU was per-position argmax over teacher-forced logits, not autoregressive decoding; v3 corrected this to beam-3.
5. *Validation mask inversion* (v1-v3): the data loader sets `txt_mask=True` at *valid* positions, but token counts (and the v2 greedy truncation length) used the negated mask, counting *padding* positions.
6. *Loss-normalisation mismatch* (v1-v3): the config specifies `translation_normalization: batch`; our code used per-token mean CE. The released log pins the original semantics exactly: its step-14 dev **Translation Loss** 43,440.80078 equals 7,813 valid dev target tokens $\times \ln(\text{PPL } 259.84015)$, i.e. token-sum CE divided by the number of sentences for the training loss. Under Adam this changes the effective loss scale by the mean target length ($\sim 15\times$), altering both the gradient-clipping regime and the relative strength of weight decay.
7. *Stopping-rule mismatch* (v1-v3): a 300-epoch cap with early-stop patience 15. The released log refutes patience-based early stopping outright: the released run improved last at validation #130 (step 1820, BLEU 13.38) and then ran *72 further non-improving validations*, ending at step 2828 with LR 1.15×10^{-5} —an external termination, not a patience trigger. The only stopping condition reproducible from the config is the learning-rate floor (`learning_rate_min=1e-8`), which requires 52 plateau decays of $0.8\times$.

The faithful re-implementation (`train_faithful.py`) corrects all seven defects and takes every remaining parameter from the released config: seed 42 (the released seed) as the primary run plus seeds 43–49 for within-implementation dispersion; joint CTC+translation with batch normalisation (both loss terms token-sum $\div$ sentences); step-level validation every 14 optimiser steps; autoregressive beam-3 selection identical to the evaluation decoder; `ReduceLROnPlateau(mode=max, factor 0.8, patience 5)` stepped once per validation—the released log’s 20 decays each fire on exactly the 6th consecutive non-improving validation, matching this wiring; best-checkpoint initialisation at $-\infty$ so the first validation counts as improved (matching the released log’s first *); a 3,000-epoch cap with the LR-floor stopping rule; and gradient clipping at 1.0 (disclosed as an author inference—the config does not specify clipping; with batch-normalised loss the clip genuinely binds, pre-clip gradient norms over the first ten steps being 377, 568, 145, 33, ..., whereas under the earlier token-mean loss it almost never bound). A 56-step smoke run reproduces the released log’s step-14 quantities to within 1.1%: dev token-sum NLL 42,981.5 vs. 43,440.8, PPL 245.0 vs. 259.8 (`results/faithful_smoke_gradnorms.json`). The one released-log column we cannot reproduce is the **Recognition Loss** magnitude (our per-sentence dev CTC is 39.9 at step 14 vs. the logged 17.28); the normalisation of the original CTC term is not identifiable from public artifacts. In practice each faithful run terminated at the plateau scheduler’s *numerical* floor (LR frozen at $4.4\text{e-}8$; torch rejects decay steps smaller than

its internal $\varepsilon=10^{-8}$), with the best checkpoint unchanged for the final 218–338 validations. The SI config-mapping table (SI Sup. F) maps every load-bearing field to the original-framework semantics, our v1–v3 semantics, and the faithful semantics.

Table 1: Implementation ladder from the original defective protocol to the faithful reimplementation. All variants share the released architecture, data, and vocabularies; they differ in loss composition, normalisation, validation cadence, selection decode, and stopping rule. Dev BLEU-4 is the uniform beam-3 full-pool value; gap is the canonical PURE–REC sacreBLEU-4 difference on the 641-item test set; readout is full-training-pool free-decode BLEU-4 (released: 78.8).

Protocol remaining)	(defects	Selection / validation	Dev BLEU-4	Readout	Gap range (neg. frac.)
Translation-only epoch / NLL (1–4, 5–7)	/	NLL / epoch	6.5–9.4	7.2–9.3	[−1.85, −0.80] (14/14)
validation-freq-misread (2 corrected)		NLL / epoch	8.3–9.8	—	[−1.70, −0.52] (4/4)
Translation-only / step-val / BLEU (1, 4, 5–7)		greedy / 14 steps	7.6–10.5	—	[−1.37, −0.36] (10/10)
Joint-loss / step-val / greedy-BLEU (4–7)		greedy / 14 steps	6.3–10.2	—	[−1.59, −0.48] (8/8)
Joint-loss / step-val / beam-3 (5–7)		beam-3 / 14 steps	8.7–10.5	—	[−1.38, −0.37] (8/8)
Faithful (all corrected; seeds 42–49)		beam-3 / 14 steps	11.59–12.62	11.45–12.08	[−2.80, −1.37] (8/8)
Released evaluator	—		13.38	78.8	+10.24

3.5 Discovery timeline and pre-registration status

The +10.24 probe response was first observed on 2026-07-24 (canonical materialization; earlier pre-exclusion gap +11.01 on 2026-07-22). The six prospectively specified legacy seeds (101–606) were specified before their results were inspected; the eight extension seeds (707–1405) were added post-hoc. The Jaccard retriever, PURE protocol, and 641-query set were locked before reconstruction training began, after exploring the probe matrix now reported in SI Sup. S. No external pre-registration exists; these six seeds constitute the only prospectively specified within-project follow-up evidence. Because the Jaccard pure-copy probe was selected as the maximal-gap probe post-hoc, its CI is descriptive, not prospectively confirmed. Full provenance in SI Sup. C.

3.6 Metrics and uncertainty

The recorded test poses (REC) are decoded by each BT recognizer; we avoid the abbreviation “ground truth” because recorded poses are inputs to a learned measurement model, not a semantic gold standard. We report corpus BLEU-4 on the 0–100 sacreBLEU scale Post (2018) (sacrebleu 2.5.1 signature:

BLEU|nrefs:1|case:mixed|eff:no|tok:13a|smooth:exp|version:2.5.1; max n-gram order 4; effective_order=False). BLEU-1, chrF, ROUGE-L, BERTScore Zhang et al. (2020), and WER (jiwer 3.1.0) provide cross-metric consistency checks (not independent semantic validation, as all share decoded outputs). BT likelihood provides a non-decoded control; pose-DTWp was withdrawn (see §4.1).

Uncertainty is quantified with donor-cluster bootstrap (primary; 10,000 resamples; clusters defined by donor identity, 565 unique donors across 641 queries, 61 donors reused with maximum reuse 6, Gini 0.11; SI Sup. G) and item-level bootstrap (SI Sup. G). Clustering sensitivity for the headline gap: donor-cluster [+8.87, +11.64] and donor two-way [+8.79, +11.74] match the item-level CI [+8.88, +11.62] to within 0.03 BLEU; signer (9 clusters) [+9.29, +11.39]; broadcast-date (297) [+8.83, +11.72]; signer×show (17) [+7.89, +11.91]; the broadcast-show split has only 2 clusters and widens to [+5.16, +11.44]—an exploratory sensitivity reference, not a confirmatory interval (full designs in SI Sup. G). A pairwise difference is called detectable when its interval excludes zero. For seed dispersion we report *descriptive* Student-*t* intervals over the 14 paper-derived seeds (six primary + eight extension) and over the 10 step-corrected seeds; these intervals assume same-implementation seed outcomes are approximately IID/normal, an assumption that leave-one-out sensitivity (SI Sup. B) checks for robustness but cannot validate. They should be read as “given a future random seed under this identical implementation” dispersion summaries, not as calibrated population-level prediction intervals or independent replication statistics; raw seed values are listed in SI Sup. B and SI Table S2. The six primary seeds give a descriptive interval [−2.48, −0.28]; all 14 leave-one-seed-out folds keep the interval below zero (SI Sup. B). The headline gap is a **post-selected descriptive estimate** (§4.1); all other contrasts are Holm-corrected within a defined family or explicitly labelled descriptive (SI Sup. C).

3.7 Terminology, claim hierarchy, and stratification

Throughout, *run* denotes a single training execution; *checkpoint* the selected weights from that run; *unique weight binary* counts distinct SHA-256 hashes (one validation-freq-misread seed-202 artifact is byte-identical to a long-schedule artifact); *family* a group of runs sharing a training protocol; *evaluator* a checkpoint used to decode fixed outputs. Claims use four evidence tiers: Descriptive (D; no formal error control), Sensitivity (S; explicit post-hoc condition), prospectively specified within-project follow-up (C; only the six legacy seeds 101–606, which share one implementation and are seed-level variation, not independent replications), and Not Identified (NI; e.g., the causal checkpoint-identity effect). Checkpoint families stratify into four tiers: *Faithful* (8 runs, seeds 42–49, all seven defects corrected, seed 42 first; primary evidence for RQ1 and the non-reproduction observation), *Partially-corrected* (14: 10 step-corrected + 4 validation-freq-misread; retain the normalisation, mask, CTC-off, and stopping-rule defects), *Secondary* (14 legacy seeds, of which the 6 prospectively specified provide the only prospectively specified non-reproduction direction), and *Diagnostic* (40: joint-loss, confirmation, rescue, distillation, ladder, long-schedule). The non-reproduction claim rests primarily on the 8 faithful runs (range [−2.80, −1.37], all negative, at test-split competence matching the released evaluator), corroborated

by the partially-corrected (step-corrected range $[-1.37, -0.36]$) and legacy families (within-implementation dispersion interval $[-2.48, -0.28]$ over the six prospective seeds; below zero in all leave-one-seed-out folds). Diagnostic runs triangulate and must not be cumulated with prospectively specified evidence. Full terminology registry and claim-hierarchy table in SI Sup. P; the auto-generated accounting table (76 trained / 74 decoded / 73 unique / 69 non-degenerate, all 69 gaps negative) is SI Table S26, generated from `results/accounting_table.json`.

4 Results

4.1 Adversarial probe response: an attack result, confirmed and localised

Under the released SLRTP2025 evaluator, source-conditioned whole-donor retrieval scored 23.02 sacreBLEU (0–100 scale), compared with 12.78 for the recorded test poses—a difference of +10.24 BLEU points. We present this as the output of a deliberately adversarial *unit test* of the released evaluator, not as an empirical finding about evaluator behaviour in deployment: the retriever selects donors by surface vocabulary overlap with the source, which is identical to the scoring reference, so the number measures the evaluator’s susceptibility to this specific shortcut rather than general retrieval quality or signing adequacy. Because the headline cell was selected as the maximal-gap cell among the explored probe cells (SI Table S46), we do not attach a confirmatory-looking interval to it; its resampling intervals are reported in SI Sup. G as post-selection descriptive quantities.

Three new results characterise the attack precisely. **(i) Frozen-split confirmation:** the identical probe construction applied to the development split (515 items, never used for probe construction or exploration) reproduces the response at consistent magnitude—REC 13.38 (matching the released training log’s 13.38 to three digits, independently validating the decode pipeline) vs. PURE 21.47, a gap of +8.10 against the test-split +10.24 (SI Sup. S). **(ii) Reference-free retrieval collapses the effect:** a pose-similarity retriever that never touches any text—donors selected by nearest-neighbour distance between query and donor pose signatures (time-averaged pose plus mean absolute velocity, z-normalised)—scores 0.45 (gap -12.33), indistinguishable from random donors (0.90, gap -11.88) and far below text-retrieval TN-PURE (23.02). The positive response is therefore entirely dependent on selecting donors with the scored reference text (SI Sup. S). **(iii) Deployment-realistic systems do not show it:** a noisy-gloss retrieval pipeline scores only -0.13 relative to REC under the same evaluator (§4.5).

Further controls localise the effect to source-conditioned donor retrieval (SI Table S32): pure donor copy (23.02) exceeded PT-composed (18.86), oracle-gloss (11.4–11.8, a leakage diagnostic), PT baseline (1.59), and 20-seed random-donor baselines (0.90/0.77). Source-text Jaccard was the strongest retriever (vs. LaBSE 19.2, multi-SBERT 16.5). Non-decoded controls do not reproduce the gap: teacher-forced NLL gives REC 3.060 vs PURE 3.479. Complementary text metrics (BLEU-1, chrF, ROUGE-L, WER, BERTScore) agree on direction (SI Table S32); decoding-parameter

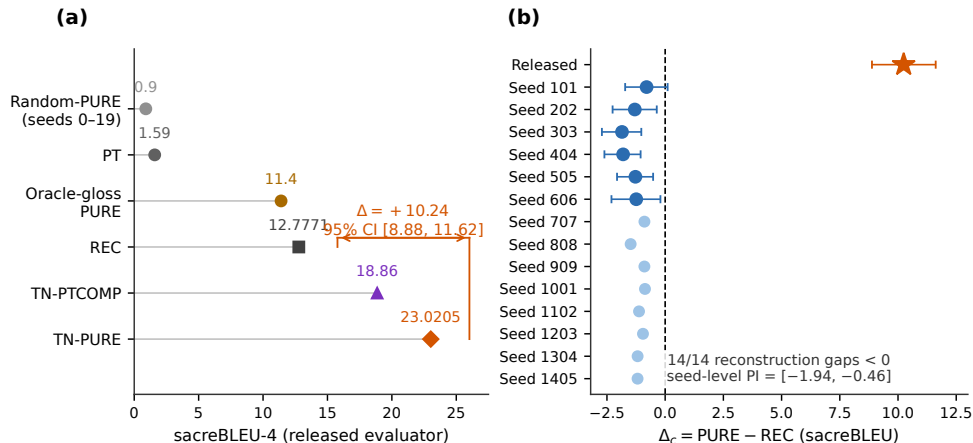


Fig. 1: Adversarial unit-test response under the released evaluator and its absence across constructible evaluators. (a) System scores under the released evaluator (sacreBLEU-4, canonical 641-query protocol): TN-PURE 23.02 vs. REC 12.78 ($\Delta = +10.24$, post-selected), with TN-PTCOMP 18.86, PT 1.59, oracle-gloss PURE 11.4, and random-donor PURE 0.90 (seeds 0–19, ± 1 SD). (b) Per-checkpoint PURE–REC gaps: released evaluator (+10.24) vs. fourteen legacy recipe reconstructions (seeds 101–1405), all 14 negative (donor-cluster bootstrap 95% CIs; descriptive seed dispersion range $[-1.85, -0.80]$). The partially-corrected, joint-loss, and faithful families (gaps $[-2.80, -0.21]$ overall; SI Sup. P accounting table) are shown in Fig. 2; none produces a positive gap, including the faithful runs whose test-split competence matches the released evaluator.

sweeps (beam 1/3/5, length penalty, max length) leave the gap at +9.9 to +10.8 (SI Sup. F).

Probe multiplicity and post-selection status.

Before freezing the headline probe, we explored four retrieval strategies crossed with three construction variants on the released evaluator alone (6 of 12 cells explored, plus two random baselines; full matrix in SI Table S46 and `results/probe_multiplicity.json`). The headline Jaccard pure-copy probe was selected as the *maximal-gap* cell; we do not apply a multiplicity correction because the family of explored probes was defined after observing the gaps, so no correction would make the result confirmatory. The six prospectively specified legacy seeds (§3.5) were pre-specified and provide the only prospectively specified evidence (for the non-reproduction direction, not for the gap magnitude).

The probe reversal motivates the prior question: can the public recipe reproduce the released evaluator at all?

4.2 RQ1: What the faithful re-implementation recovers, and what it does not

Table 2 reports the faithful family (eight seeds, §3.4) against the released evaluator on four dimensions: dev competence, test-split decoding of the recorded poses (the probe panel’s own REC baseline), full-training-pool readout, and the probe gap. Three findings follow.

Table 2: The faithful family (`train_faithful.py`, seeds 42–49; all seven defects corrected) versus the released evaluator. Dev/test BLEU-4 are uniform beam-3 decodes (515/641 items); readout is full-training-pool free-decode BLEU-4 (7,060 items) with exact-match rate; gap is the canonical PURE–REC sacreBLEU-4 difference on the 641-item test panel. The released evaluator’s row uses the same protocols.

Seed	Dev BLEU-4	Test REC	Train readout (EM%)	PURE	Gap
42 (released seed)	12.05	13.11	11.96 (2.4%)	10.32	−2.80
43	11.59	11.60	11.45 (2.3%)	10.22	−1.37
44	11.95	12.39	11.74 (2.3%)	10.13	−2.25
45	12.62	12.04	11.87 (2.5%)	10.08	−1.97
46	11.80	12.26	11.48 (2.3%)	10.16	−2.10
47	12.03	12.53	12.08 (2.4%)	10.34	−2.20
48	12.17	12.33	11.67 (2.4%)	9.98	−2.35
49	11.93	11.86	11.48 (2.4%)	9.77	−2.09
Released evaluator	13.38	12.78	78.8 (70.7%)	23.02	+10.24

Finding 1 (surface competence largely recovered).

The faithful runs reach dev BLEU-4 11.59–12.62 (primary seed-42 run: 12.05) against the released 13.38—closing most of the gap that the seven defective implementations left open (their ceiling: 10.5). On the test split, where the probe panel itself is scored, the faithful evaluators decode the recorded poses at 11.60–13.11: the seed-42 run *exceeds* the released evaluator’s 12.78, and five of eight seeds are within 0.8 BLEU of it. The author-operational competence gate (dev ≥ 12.38 ; §3.1) is met by one seed (45). Under any reasonable reading, the earlier failure to approach the released evaluator’s competence is explained by the seven implementation defects, not by the public artifacts.

Finding 2 (training-pool readout signature not recovered).

What the faithful family does not reproduce is the released evaluator’s training-pool behaviour: full-pool free-decode readout is 11.45–12.08 BLEU with 2.3–2.5% exact match, against the released 78.8 BLEU / 70.7% EM. The residual gap is therefore concentrated in a *memorisation signature*—near-perfect decoding of the evaluator’s own training pool at retained in-domain generalisation—and not in surface competence. Five leakage controls (SI Sup. M) rule out teacher-forcing, output–reference misalignment, pose-ignorant lookup, and narrow-text memorisation shortcuts for the released

evaluator’s readout, so the signature reflects pose-conditioned training-pool exposure that no constructible run, faithful or defective, approaches.

Finding 3 (probe response tracks the signature, not surface competence).

All eight faithful evaluators give negative probe gaps (−2.80 to −1.37; Table 2), including the seed-42 and seed-47 runs whose test REC (13.11, 12.53) matches or exceeds the released evaluator’s 12.78. The non-reproduction of the adversarial probe response therefore cannot be attributed to uniformly lower surface competence. Across the 69 non-degenerate constructible evaluators (gap range [−2.80, −0.21]; SI Sup. P), the probe response instead tracks the training-pool signature that only the released evaluator exhibits.

Descriptive triangulation for the constructible family as a whole (Fig. 2b,c; SI Sup. D): across the 39 checkpoints with uniform full-pool readouts (38 constructible plus released), training-pool readout tracks decoded competence within the constructible family (Spearman $\rho=0.916$; $\rho=0.907$ with the released evaluator excluded, confirming the correlation is not driven by the released extreme point) but is uncorrelated with the PURE-REC gap ($\rho=-0.082$); the released evaluator remains the only checkpoint combining high readout with retained generalisation, and no distillation student (15 runs, SI Table S36) transfers the readout pattern. The reversal is locally robust near the released checkpoint (Gaussian weight noise $\sigma \leq 3 \times 10^{-3}$; nine released-weight fine-tunes), but those probes test the released weights’ vicinity, not an independent construction.

Table 3: Summary of the three primary descriptive comparisons over the full constructible family. Family n : (i) 74 beam-3-decoded training runs (range over 69 non-degenerate; 5 degenerate excluded); (ii) 38 constructible checkpoints with uniform full-pool readout (released listed separately); (iii) gate-eligible non-distillation training runs. Per-view family composition in SI Sup. D.

Comparison	Family n	Released evaluator	Family range	Released – family max
(i) Gap-competence	74	+10.24 at dev 13.38	[−2.80, −0.21] (69 non-deg.)	+10.45
(ii) Train-pool readout BLEU	38	78.8 (70.7% EM)	6.6–12.08 (1–2.5% EM)	+66.7
(iii) Competence gate (dev)	60	13.38	6.5–12.62 (faithful max 12.62; 1/8 pass)	+0.76

Leakage sanity checks for the train-pool readout.

The 78.8/70.7% readout is not a trivial implementation artifact: five controls (SI Sup. M) rule out four shortcut classes — teacher forcing, output-reference misalignment (label permutation collapses BLEU from 78.6 to ~ 1.0), pose-ignorant ID/cache lookup (input-side pose permutation collapses BLEU to <1 ; a same-signer

held-out control reaches only 12.19 BLEU / 3.2% EM), and narrow-text memorization. The checks do not identify the training mechanism but are consistent with pose-conditioned training-pool exposure.

Distillation and competence extrapolation.

Fifteen distillation students ($\alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}$, three seeds each) all have negative gaps and do not transfer the readout pattern (SI Table S36). Competence extrapolation (OLS/GP) and per-item gap decomposition are reported in SI Sup. K; with the faithful family the residual unobserved dev interval shrinks to (12.62, 13.38), and no constructible checkpoint enters it.

Synthesis.

The released evaluator is the only checkpoint combining high training-pool readout (78.8 BLEU, 70.7% EM) with retained in-domain performance (13.38/12.78). No constructible run, faithful or defective, reproduces either the +10.24 probe response or this combination; the faithful family closes the surface-competence dimension but not the signature.

4.3 Descriptive observation: probe non-reproduction across 69 constructible evaluators

This subsection reports a *descriptive observation*, not a research question. Across every constructible evaluator—the eight faithful runs, fourteen partially-corrected runs, fourteen legacy seeds, and the diagnostic families—the probe ordering does not reproduce: all 69 non-degenerate decoded runs have strictly negative PURE-REC gaps (range $[-2.80, -0.21]$; 0 positive). For the fourteen legacy reconstructions, the within-implementation seed dispersion interval is $[-1.94, -0.46]$ (mean -1.20 , SD 0.33 ; the six prospectively specified seeds alone give $[-2.48, -0.28]$; below zero in all fourteen leave-one-seed-out folds), and for the six prospectively specified seeds the original-minus-run difference-in-differences estimates range from $+10.89$ to $+11.99$, every interval excluding zero. The faithful family shows the same sign at substantially higher competence ($[-2.80, -1.37]$; Table 2). Because the faithful runs match the released evaluator’s *test-split* decoding competence while lacking its training-pool signature, the non-reproduction is no longer attributable to surface competence at all; what remains confounded is the training-pool memorisation signature with any further unrecovered training-mechanism difference. Per-seed values for the legacy family are tabulated in SI Table S2; competence is assessed on the development split, never on the probe panel.

Competence gate and trajectory analysis.

A checkpoint is competence-matched only if both dev BLEU-4 ≥ 12.38 and dev WER ≤ 86.37 (one-sided tolerance around released 13.38 / 83.37). Of the 60 gate-eligible non-distillation training runs, exactly one passes: faithful seed 45 (dev 12.62). The 9 released-weight fine-tunes (SI Sup. F) all pass (dev 12.92–13.18), but they perturb

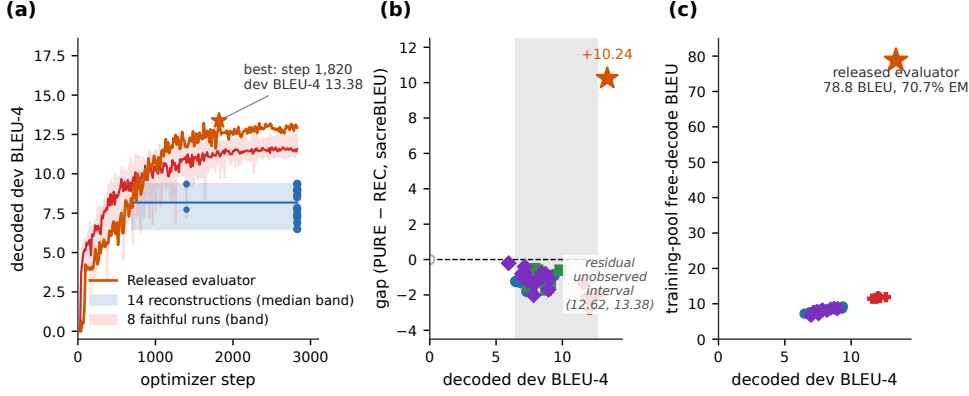


Fig. 2: Decoded competence and training-pool readout separate the released evaluator from every constructible checkpoint. (a) Dev-BLEU-4 training trajectories on a common optimizer-step axis: the released evaluator (orange) reaches 13.38 at step 1,820 of 2,828 (archived validation log, 202 logged points); fourteen defective-protocol reconstructions form a median band (blue) and the eight faithful runs a second band (red) that tracks the released trajectory closely before plateauing at 11.6–12.6. (b) Canonical PURE–REC gap vs. decoded dev BLEU-4 by family: reconstructions (circles), validation-freq-misread/step-corrected/confirmation/long-schedule (squares), joint-loss greedy/beam-3 (triangles-up/triangles-down), rescue (triangle-right), distillation students (diamonds), released evaluator (star); the gray band spans the constructible 6.5–12.6 range (uniform protocol; faithful runs in red plus-signs); the residual (12.62, 13.38) interval is unobserved by recipe-constructed checkpoints (the nine released-weight fine-tunes occupy 12.92–13.18 but are weight perturbations); degenerate checkpoints are hollow gray points. (c) Full-training-pool free-decode readout (beam-3, full 7,060 poses) vs. decoded dev BLEU-4: every constructible checkpoint reads out at most 12.1 BLEU (defective families 6.6–9.3; faithful 11.5–12.1), while the released evaluator reads out 78.8 BLEU (70.7% exact match) at dev 13.38. Families are descriptive; checkpoints are not IID replicates.

the released weights rather than constructing from the recipe. The gate conclusion is threshold-insensitive across a $\pm[0.5, 2.0]$ BLEU $\times$ $\pm[1.5, 4.5]$ WER grid (SI Sup. D–E).

What the released training log establishes.

The bundle’s `validations.txt` logs dev BLEU-4 reaching 13.38 at step 1,820 (Fig. 2a); the defective families plateau at 8–10 and the faithful family at 11.6–12.6, so the final 0.76-BLEU ascent and the readout signature are not explained by public artifacts (§4.2). Input degradation (temporal downsampling) compresses both scores but does not eliminate the reversal (at $k=8$: REC 8.75, PURE 11.69, gap +2.94; SI Sup. F).

Cross-metric consistency (SI Table S32).

Under the original evaluator, the PURE-REC gap is positive and large for all decoded-text metrics (BLEU-1 +20.94, chrF +12.93, ROUGE-L +17.87, WER +13.90 sign-flipped so positive means PURE has lower error than REC; sign conventions in SI Table S32 footnote), while teacher-forced NLL is more negative for PURE (−0.419, also sign-flipped), consistent with the likelihood-vs-decoded-metric divergence Jiang et al. (2025). Under all six prospectively specified legacy reconstructions the gap is negative across every metric. The probe response is therefore a property of decoded-text metrics under the original evaluator, not specific to sacreBLEU-4. Full table in SI Table S32.

Rank stability (SI Sup. H).

On a fixed nine-system stress-test set, the released evaluator ranks TN-PURE first; five of six reconstructions rank REC first, with moderate cross-checkpoint agreement (mean $\tau_b=0.44$, $\rho=0.57$).

Reference-frame sensitivity: matched-subset analysis.

Two 2026 corpus audits bear directly on this reversal. Czehmann et al. Czehmann et al. (2026) released manual sign-to-text back-translations of the test set (one deaf fluent L2 DGS signer; per-item confidence flags), documenting information loss, lexical errors, and translationese in the original speech-transcribed references, and showed that reference replacement already changes BLEU scores and rankings on PHOENIX-2014T. Alkain et al. Alkain et al. (2026) quantified train-test text overlap across sign-language translation corpora, showing that a small fraction of highly training-like test items disproportionately inflates corpus BLEU. We reproduce the overlap pattern locally: the 32 test items whose reference exactly matches a training text score REC 40.26 / PURE 78.73 (gap +38.5), while the remaining 609 items score REC 11.93 / PURE 22.20 (gap +10.3).

Czehmann et al. compared original- and human-reference conditions on the same confidence=1 subset (462 of 642 items); we follow the same protocol. On the matched 461-item subset (after join with our 641 released test IDs), the original-reference gap is +10.03 BLEU points (REC 14.94 / PURE 24.98) and the human-reference gap is +0.95 (REC 7.35 / PURE 8.30): a *reference-frame intervention contrast* of 9.08 BLEU points (a 90.5% gap reduction under this intervention; bootstrap CI [7.44, 10.86] on the absolute contrast and [0.81, 1.01] on the ratio; SI Table S41; SI Fig. S5; per-item paired view in SI Fig. S2). The interval for the residual gap includes zero (paired item CI [−0.13, 1.99]). This contrast is a *single-annotator, 461-item, post-hoc point estimate*; it is an intervention contrast, not a reference-defect effect: reference replacement simultaneously changes (i) reference content, (ii) expression style/translationese, (iii) BLEU floor level, and (iv) the lexical coupling between retrieval selection and scoring reference. Teacher-forced BT likelihood does not reproduce the probe response under either reference (REC NLL 3.060 vs PURE 3.479 on original refs; 3.41 vs 3.71 on human refs) Jiang et al. (2025).

Scoring against both references jointly leaves the gap at +11.50, *larger* than original: adding the original reference injects a second lexically coupled target. The

probe response is thus specific to reference frames containing the original speech-transcribed register. This is a descriptive contrast, not a causal attribution. Under all six reconstruction checkpoints the human-reference gap is negative (-0.15 to -0.65), and the asymmetry is consistent across metrics (chrF 82.2%, BLEU-1 84.0%; SI Table S46). The asymmetric PURE-vs-REC drop (under reference replacement REC drops 50.8%, PURE drops 66.8%, a $1.31\times$ relative-drop ratio; the absolute-drop ratio is $16.68/7.59 = 2.20$) is numerically inconsistent with simple symmetric floor compression (SI Table S46), though BLEU’s non-linearity precludes a definitive test. The human back-translations are sign-conditioned paraphrases of the recorded test poses, not judgments of the replay donors, so semantic adequacy of TN-PURE-v1 is left unassessed.

4.4 Donor-origin contrast

If the released-evaluator probe response requires exposure to training-pool donors, retrieval restricted to unseen poses should remove or invert it: under the original evaluator, UNSEEN-PURE-v1 (retrieving from the 640 other test poses) scores 8.51 sacreBLEU vs. REC 12.78 (gap -4.27), a donor-pool substitution estimand of $+14.51$ BLEU points (SI Table S40). This is a *pool-substitution difference*, not a pure exposure contrast: SEEN and UNSEEN differ simultaneously in pool size (7,060 vs 640), median donor-reference Jaccard (0.50 vs 0.37), donor reuse (Gini 0.11 vs 0.29), and training-pool exposure.

Matched donor-pool analysis (SMD<0.1 control).

Restricting to the 148 items where seen-donor and unseen-donor Jaccard similarities differ by at most 0.05 (SMD 0.065 vs. $+0.554$ on the full 641), the pool-origin contrast is $+9.56$ BLEU [$+7.47, +11.76$]. The change from the unmatched $+14.51$ is not a percentage attribution—the two estimates target different populations and the caliper shifts length, density, and difficulty distributions. This is a Jaccard-balanced association, not an identified training-exposure effect (pool size and donor diversity remain uncontrolled; `results/matched_donor_pool.json`, SI Sup. E).

Pool-size-controlled resampling.

Equalling pool size directly—repeatedly subsampling 640-item donor pools from the 7,060-item training pool, applying the same retrieval rule, and scoring the released evaluator’s cached hypotheses—gives, over 1,000 resamples, a pool-origin contrast of $+8.38$ BLEU (95% conditional simulation interval [$+7.44, +9.34$], fraction positive 1.00; `results/donor_pool_resampling.json`). Pool size and donor diversity account for part of the unmatched $+14.51$, but a substantial contrast persists at equal pool size; it remains an association, not an identified exposure effect.

Balanced factorial and path decomposition.

The 605-query factorial (matching seen/unseen donors at high/low LaBSE similarity; SMD $+0.4$ within strata) estimates the pool-origin main effect at $+2.08$ [$+1.56, +2.57$], similarity at $+6.96$ [$+5.93, +8.08$], and interaction at $+3.70$ [$2.63, +4.79$] (Holm

$p=0.0003$); across reconstructions the seen main effect shrinks to $+0.51$. Because seen donors remain more similar than unseen within strata, this is a provenance contrast, not an identified training-exposure effect. The matched-subset analysis above provides a *differently conditioned association* (caliper-restricted $n=148$ vs. factorial $n=605$ vs. full $n=641$, three different target populations); the factorial is retained for its 2×2 decomposition.

4.5 Competitive-system context

Because the audit’s adversarial probe is constructed rather than competitive, the public SLRTP2025 leaderboard offers external context: the “Ground Truth” row reproduces our canonical REC numbers exactly (BLEU-4 12.78, Walsh et al. (2025); SI Table S35). The retrieval-based winner USTC-MoE sits roughly even with REC on lexical BT metrics (BLEU-4 -0.72 ; SI Table S35). We re-decoded two realistic retrieval outputs under all available evaluators (SI Sup. P): the deployment-realistic noisy-gloss retrieval pipeline decodes at 12.65 under the released evaluator (gap -0.13 vs. REC), and our rebuild of the winner’s fixed outputs from its released intermediates decodes at 2.42 (gap -9.94 ; a failed reproduction, not representative of the original entry). Under all 70 non-degenerate reconstructed evaluators, including the eight faithful runs, the noisy-gloss system gives negative gaps throughout (range $[-3.22, -0.07]$; faithful sub-range $[-3.22, -1.59]$), and USTC-MoE remains deeply negative ($[-9.52, -4.28]$). Neither realistic system recovers a retrieval-favoring ordering under any evaluator. The $+10.24$ gap is specific to the constructed adversarial probe, and we do not claim that real competition rankings would change across checkpoints. Archiving fixed per-team outputs alongside public scores is the most direct suggestion we can offer for enabling rank-stability audits.

Per-item gap decomposition (experiment C, SI Sup. K).

An OLS regression ($R^2 = 0.33$, $n = 641$) attributes the largest share of per-item gap variance to high-overlap-train count, template density, and donor-reference Levenshtein distance; removing the signer block changes R^2 by $\leq 1\%$ (ΔR^2). The response variable is a sentence-level smoothed-BLEU proxy: the per-item PURE-minus-REC difference computed by aggregating each item’s n-gram sufficient statistics under the corpus exp-smoothing convention. This proxy is *not* an additive decomposition of the non-linear corpus BLEU (the mean of the per-item proxies does not equal the corpus difference), so the regression explains the sentence-level proxy response, not exact corpus contributions; SEs are heteroskedasticity-robust (HC3) and also reported with two-way clustering on signer and broadcast show as a sensitivity check. The model reported is the AIC-minimal specification across a small grid of additive and pairwise-interaction candidate regressors selected before fitting; we do not claim it is the unique correct model. Full coefficient table, alternative-model sensitivity, and figure in SI Sup. K.

5 Discussion

The audit yields a sharpened picture of what the SLRTP2025 public artifacts reproduce, organised around RQ1 (reconstruction), RQ2 (reference-frame and donor-origin sensitivity), and one descriptive observation (probe non-reproduction).

RQ1—Competence largely recovered; the training-pool signature is the residual gap. After seven implementation defects were identified and corrected (§3.4), the faithful family reaches dev BLEU-4 11.59–12.62 against the released 13.38, with test-split decoding up to 13.11 against the released 12.78. The earlier “failed reconstruction” competence gap is therefore attributable predominantly to implementation defects in our own reconstruction code—a result that itself illustrates how easily an eval-only release invites exactly such defects (four of the seven were invisible without forensic arithmetic against the released training log). What no constructible run reproduces is the released evaluator’s *training-pool signature*: full-pool readout 78.8 BLEU / 70.7% exact match against a constructible ceiling of 12.08 / 2.5%. The residual unobserved dev interval shrinks from (10.5, 13.38) to (12.62, 13.38), and the source of the signature—longer exposure, a different data-order or augmentation regime, or an undocumented mechanism—remains unidentified from public artifacts.

Descriptive observation—probe non-reproduction now separated from surface competence. All 69 non-degenerate constructible evaluators give negative probe gaps (range $[-2.80, -0.21]$). Because the faithful runs match or exceed the released evaluator’s test-split decoding competence while still lacking the signature, the non-reproduction can no longer be attributed to uniformly lower surface competence; within everything we can construct, the probe response tracks the training-pool signature that only the released evaluator exhibits. This remains a descriptive association, not a causal claim: the signature and any further unrecovered training-mechanism difference are confounded, and only an independent construction that reproduces the signature could separate them.

RQ2—Reference-frame and donor-origin sensitivity. On the matched 461-item confidence=1 subset of Czehmann et al.’s Czehmann et al. (2026) human back-translations, reference replacement is associated with a gap reduction to +0.95 from +10.03—a single-annotator reference-frame intervention contrast that cannot isolate reference-text defects from lexical-coupling disruption, paraphrase shift, or BLEU floor effects. The matched donor-pool analysis (§4.4) shows the pool-origin contrast persists at +9.56 under SMD<0.1 Jaccard matching; the pool-size-controlled resampling (640-vs-640, §4.4) gives +8.38 (conditional simulation interval $[+7.44, +9.34]$). All three pool-origin estimates remain associations, not identified training-exposure effects.

Relation to prior work.

These findings extend prior critiques of BT validity from output faithfulness to public-artifact reproducibility: improved BT scores need not imply target conditioning Hong and Košecká, Jana (2026), low recorded-pose BT baselines are documented Walsh et al. (2024), and BT likelihood is a more stable correlate of human judgment than decoded BLEU Jiang et al. (2025). Our reversal is a *decoded-sacreBLEU* reversal — teacher-forced BT likelihood does not reproduce it (REC NLL 3.060 vs PURE 3.479) — and whole-donor replay is a constructed probe, whereas SOKE-style word-level

retrieval can improve pose quality under a different representation Zuo et al. (2025). Chen et al.’s Chen et al. (2022) reproduction failure of a released metric had the same anatomy we document here for a full evaluator: an eval-only release, underspecified training semantics, and reconstruction attempts that failed until the semantics were pinned by forensic comparison with the released artifacts.

Scope limitation: our reconstruction vs. public-artifact sufficiency.

Our conclusion is strictly about what our disclosed implementations construct from the public artifacts. The faithful family falls 0.76 dev BLEU and a factor of ~ 6.5 in training-pool readout short of the released evaluator; the residual may reflect artifact incompleteness or a further inference error of the same kind as the seven we corrected. The artifacts *were* sufficient to recover most surface competence; the sufficiency question is resolved only in that operational sense.

Value, finding, and conclusion reproducibility.

Following the tripartite framework of Cohen et al. (2018): *Value* (recomputing the headline numbers from the released artifacts) is reproduced—REC 12.78, PURE 23.02, gap +10.24 regenerate from the canonical donor registry. *Findings* split: the released evaluator’s competence is now substantially value-reproducible (faithful family 11.59–12.62 dev), while its training-pool readout signature and probe response are not reproduced under any of the 69 constructible non-degenerate checkpoints. *Conclusions* about artifact sufficiency must be stated dimension-by-dimension, as above.

Actionable recommendations for benchmark organizers.

The audit yields three transferable lessons, scoped to the constructed replay-probe stress case on PHX-public rather than to general SLP leaderboard stability.

Lesson 1: a released evaluator is a research artifact, not an infrastructure component. Seven implementation defects—several pin-able only by arithmetic against the released training log—separated our initial reconstructions from the released competence, and the readout signature could not be localised to a training stage because intermediate checkpoints are unpublished. Benchmarks should release the exact training command and seeds, pipeline scripts beyond the manifest, a SHA-256 manifest of training tensors, intermediate checkpoints, and the fixed per-team pose outputs.

Lesson 2: on template-dense benchmarks, retrieval entries may benefit from donor-exclusion disclosure. A donor-exclusion sweep is cheap and exposes the reference-coupling gap. The training-pool readout ratio (released ≈ 5.9 ; constructible 0.93–1.55) and input-side pose permutation with output-equivariance verification (SI Sup. M) are two informative descriptive probes; generalisation to other benchmarks is untested.

Lesson 3: reference-coupling artifacts warrant a supplementary reference check. The reference-frame intervention contrast under Czehmann et al. Czehmann et al. (2026) back-translations (90.5% gap reduction) is a single-annotator case study and should not be generalised beyond this signer. Template-dense benchmarks could adopt human back-translations as a supplementary reference set and flag systems whose gap shifts between frames.

Limitations.

The audit is confined to PHX-public, a template-dense weather domain with documented reference-validity and train–test-overlap issues Czehmann et al. (2026); Alkain et al. (2026). Per-team pose outputs are not archived; our cross-evaluator studies decode at or below REC under all 69 non-degenerate evaluators. The faithful family terminates at the plateau scheduler’s numerical LR floor ($4.4\text{e-}8$) rather than the configured floor ($1\text{e-}8$), with best checkpoints unchanged for the final 218–338 validations; the released run itself ended mid-schedule at 101 epochs, so the two stop points are not identical. The residual dev-competence interval (12.62, 13.38) is entered by no constructible checkpoint. The human back-translations are a single-annotator resource without second-annotator adjudication; the reference-frame intervention contrast is conditional on one signer’s interpretation (qualitative direction stable across confidence tiers; SI Table S41). The probe response is a decoded-BT-text-metric effect, not a sign-language quality judgment; without target-conditioned DGS signer evaluation, semantic adequacy of the donor motion is unassessed—“evaluator readability” and “signing quality” are distinct constructs, and this audit measures only the former.

6 Conclusion

This audit reports what a config-exact reconstruction of the released SLRTP2025 BT evaluator from public artifacts does and does not reproduce. Correcting seven defects in our earlier reconstruction implementations recovered most of the evaluator’s surface competence (dev BLEU-4 11.59–12.62 vs. 13.38; test decoding up to 13.11 vs. 12.78)—the earlier failure was ours, not the artifacts’. Two properties remain unreproduced across 69 non-degenerate constructible evaluators: the released evaluator’s training-pool memorisation signature (78.8 BLEU / 70.7% exact match vs. a constructible ceiling of 12.08 / 2.5%) and its response to a constructed adversarial probe (+10.24 vs. $[-2.80, -0.21]$), the latter now shown to track the signature rather than surface competence. The probe itself behaves as an adversarial unit test: confirmed on a never-explored split (+8.10), entirely dependent on reference-text-based donor selection (reference-free pose retrieval collapses to -12.33), and absent for deployment-realistic systems. The audit’s lasting contributions are the forensic method (pinning released-config semantics by arithmetic against the released training log), the provenance, checkpoint-registry, and evaluator-audit infrastructure, and the dimension-by-dimension reproducibility accounting, transferable to future learned-evaluator audits.

Statements and Declarations

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Conflict of interest.

The authors declare no competing interests.

Data governance and ethics.

The datasets and model weights are available under research-specific terms: PHOENIX-2014T under its RWTH Aachen academic licence; the SLRTP2025 pose bundle and released BT evaluator under challenge terms; the Czehmann et al. Czehmann et al. (2026) back-translations under CC BY-NC-SA 4.0; CSL-Daily under provider-specific academic terms. The artifact repository is a licence composition: the root MIT Licence covers only the RCP audit software; per-directory LICENSE/NOTICE files document the remainder (including ShareAlike obligations for the Czehmann CSVs). Pose tensors retain signer identity and biometric features; we release per-item decoded text (low re-identification risk) and a curated subset of trained checkpoints, but not per-signer raw poses. The original consent forms are not publicly available, so we cannot independently confirm that informant consent covers derivative replay-probe analysis; no new human-subjects data was collected. No Deaf/DGS stakeholders were involved in formulating the research questions or interpreting results: the adversarial-probe framing, reference-frame analysis, and recommendations reflect a hearing-researcher perspective on a DGS resource, and Deaf-led sign-language AI research norms should guide any follow-up Desai et al. (2024). The Czehmann et al. back-translations we build upon were produced by a Deaf fluent L2 DGS signer.

Data availability.

All numerical values, per-checkpoint data, decoded hypotheses, and analysis scripts are in the artifact at <https://anonymous.4open.science/r/rcp-audit-artifact-B314/README.md>. The artifact is version-controlled (git commit `a2741b1`, 2026-08-18); a claim-to-file manifest mapping each headline number to its source file and verification command is at `claim_manifest.json` (artifact root), and a single-command consistency check (`make check-consistency`, invoking `scripts/check_paper_consistency.py`) verifies the paper’s counts against the disk-scanned registry. The L1 audit environment is captured by `requirements-audit.txt` (4 packages: `numpy`, `scipy`, `sacrebleu`, `pyyaml`; install via `make install-audit`); the full `requirements.lock.txt` (236 packages) is needed only for L3 re-decoding and L4 from-scratch retraining. A persistent DOI-bearing Zenodo release is deposited at <https://doi.org/10.5281/zenodo.21987786>, available at submission and not deferred to acceptance. The supplementary material (`supplementary.pdf`) contains all SI appendices. Upstream datasets are available under research-specific terms documented in the artifact README and per-directory LICENSE/NOTICE files (PHOENIX-2014T under RWTH Aachen academic licence; Czehmann et al. human back-translations under CC BY-NC-SA 4.0, redistributed with attribution under those terms; CSL-Daily and SLRTP2025 under provider-specific academic terms).

Author contribution.

Both authors contributed to the study conception, design, data analysis, and manuscript preparation, and both read and approved the final manuscript. Specific contributions (CRediT taxonomy): **Xin Ji**: conceptualisation, methodology,

software, investigation, formal analysis, data curation, writing—original draft, visualisation. **Cheng Zhang**: methodology, supervision, writing—review and editing, project administration.

Supplementary Information (SI)

The following supplementary materials, referenced throughout the main text, are available as a separate file (**supplementary.pdf**) and in the anonymous artifact. Each appendix is annotated with the question it answers. **Sup. A** Missing Test ID Sensitivity; **Sup. B** Per-seed Legacy and Extension Cells; **Sup. C** Analysis Provenance and Multiplicity; **Sup. D** Canonical Checkpoint Registry; **Sup. E** Donor-Pool Substitution Decomposition; **Sup. F** Config-Faithful and Perturbation Details (incl. the four-column config-mapping table); **Sup. G** Cluster-Aware Bootstrap; **Sup. H** Supplementary Oracle Diagnostics; **Sup. I** Template-Density, Holdout Boundary, Donor-Exclusion; **Sup. J** Transcript-Slot Diagnostic; **Sup. K** Competence Extrapolation and Per-Item Gap Decomposition; **Sup. L** CSL-Daily Boundary Control; **Sup. M** Leakage Sanity Checks; **Sup. N** Pre-Exclusion Materialization Sensitivity; **Sup. O** BLEURT-Substitute Validation; **Sup. P** Main-Text Tables (moved from the main text); **Sup. Q** Additional Protocol Details; **Sup. R** Reconstruction Sanity Checks; **Sup. S** Frozen-Split Confirmation and Reference-Free Retrieval (incl. the full probe-multiplicity matrix). Each appendix is annotated with the question it answers; labels in **supplementary.pdf** match the references throughout the main text.

Full checkpoint registry in SI Sup. D.

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